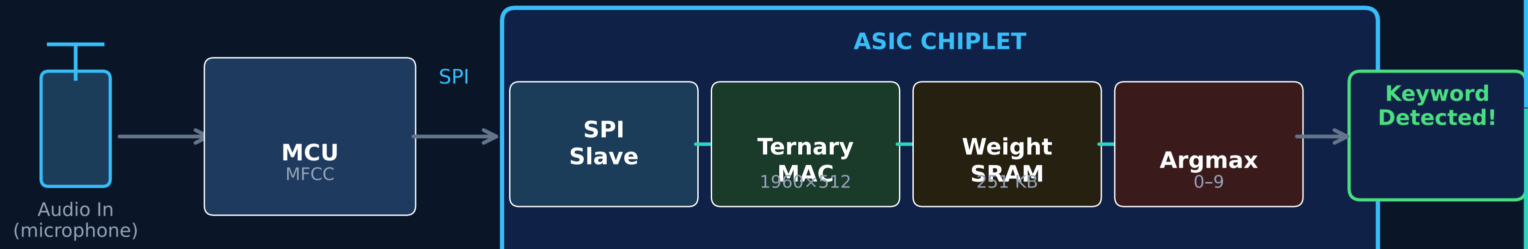
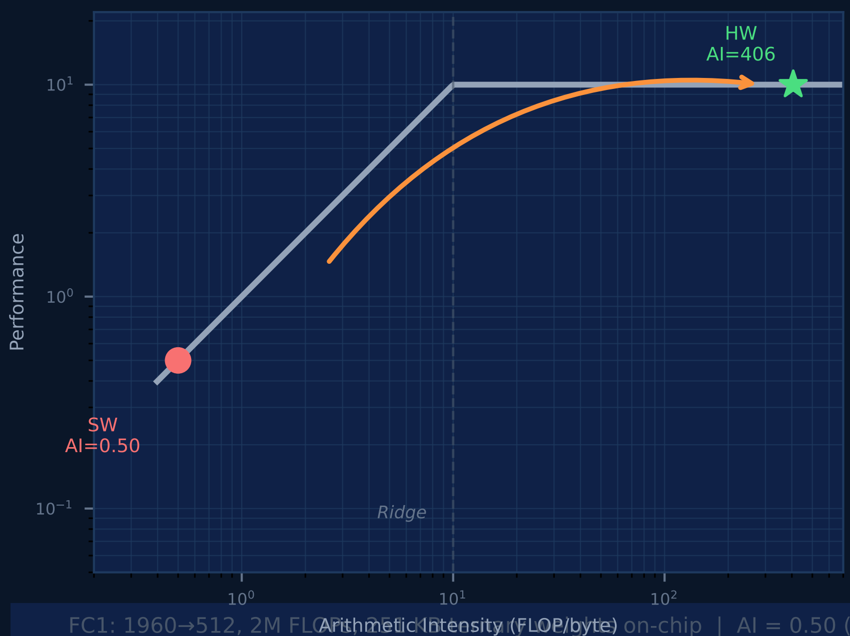
TERNARY KWS ACCELERATOR

Always-On Keyword Spotting on Silicon · ECE 410/510 · Spring 2026 · Manaf Alali

SYSTEM OVERVIEW



ROOFLINE ANALYSIS



406×
FLOP/byte on-chip

251 KB
ternary weights

>100×
HW vs SW speedup

≥92%
KWS accuracy

1 What Am I Doing?

- Custom chip: hears a spoken keyword, always-on, battery-powered
- Weights are only −1, 0, or +1 — no multiplication needed

2 How's It Done?

- Standard approach: CPU/GPU — accurate but power-hungry
- Key insight: keep weights on-chip, avoid slow external memory

3 What Am I Doing Differently?

- On-chip storage flips the bottleneck: memory → compute (see roofline)
- Weights ±1/0 replace all multipliers with add or subtract

4 Accomplished So Far

- SW: 0.44 ms/inference, confirmed bottlenecked by memory
- Math core + SPI interface written and working in simulation

5 What's Next?

- Run synthesis → real chip area, power & speed numbers
- Compare HW vs SW — target more than 100× faster

FC1: 1960→512, 2M FL[illegible] on-chip | AI = 0.50 (SW) → 406 FLOP/byte (HW) | github.com/alali1o/ECE-410